

Semantic Technology and Anthropomorphism: Exploring the Impacts of Voice Assistant Personality on User Trust, Perceived Risk, and Attitude

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ABSTRACT

A long-standing debate exists on whether robots need personality. With voice assistants (VAs) (i.e., Google Assistant and Apple's Siri) as the research context, this study employed the stimulus-organism-response (SOR) model and the theory of reasoned action to investigate how the personalities that VAs display (i.e., humanlike traits and behavior traits) influence perceived risk, perceived enjoyment, trust, attitude to use, and continued usage intention. The results show that when VAs have more humanlike linguistic traits, such as tone and phrasing, and more positive behavior traits, such as politeness and helpfulness, users enjoy using VAs more, have more trust in VAs, and display a greater willingness to continue using VAs. Unlike past studies focusing on technical aspects, the results of this study provide decision-makers with a new perspective, showing that using more humanlike designs and giving VAs unique personalities can build user trust and increase willingness to use VAs.

KEYWORDS

Attitude Towards Use, Personality, Trust, Voice Assistants

INTRODUCTION

Voice assistants (VAs) are an artificially intelligent service software that aids users in performing tasks such as querying information or purchasing goods through verbal commands and responses (Ki et al., 2020). The core technology of VAs is semantic analysis, which allows them to interpret human speech. As this technology has advanced, VAs' comprehension of human instructions has improved significantly, expanding from the interpretation of simple semantics to the emotions contained in human instructions (Du et al., 2021; Ma et al., 2022). This increase in VA comprehension has contributed to the increase in VA usage. Indeed, the total transaction amount of VA smart-home devices around the globe is projected to increase from \$22 billion in 2020 to \$164 billion in 2025, representing a growth rate of over 630% in five years (Holst, 2021). Note that countries around the world are

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continually developing voice technology applications. For example, VAs are used by nutritionists in Thailand to provide biweekly meal plans (Namahoot et al., 2021). VAs are used in Russia to learn foreign languages (Al-Kaisi et al., 2021). Other well-known retail voice robots include Amazon's Alexa, Google Assistant, and Apple's Siri. Previous studies have reported that people use VAs mainly to seek information and make purchases, and for entertainment purposes (e.g., listening to music) (McLean & Osei-Frimpong, 2019; Purington et al., 2017; Rzepka, 2019; Sciuto et al., 2018). Studies have also found that consumers like to use VAs because they are convenient, easy, and enjoyable to use (Fernandes & Oliveira, 2021; Rzepka, 2019). Enjoyment is a particularly important aspect; studies on online shopping environments indicate that consumers who do not experience enjoyment during their shopping encounters are unlikely to use the service again (Martin et al., 2015). As it is projected that over eight billion VAs will be in use worldwide by 2023 and over 51% of users will apply them to online shopping (Laricchia, 2022), it is important to understand the role of enjoyment in the customer experience.

Despite their initial success on the market, many consumers still have concerns regarding VAs. Reasons include the unfamiliarity of talking with a machine and the fact that a machine is without emotions, as well as concerns over security and privacy (McCormick, 2017; Schwartz, 2019). To enhance the service quality of VAs, researchers have suggested giving VAs unique personalities (Zhou et al., 2019). A personality refers to the thinking patterns, emotions, behaviors, and psychological mechanisms that are unique to an individual (e.g., human, animal, or robot) (Funder, 2012). Personality is a controversial topic in the development of virtual agents because "it is not easy to explain why a robot should need a personality" (Moffat, 1997). However, researchers have found that using text, images, videos, and voice to anthropomorphize virtual agents can effectively improve the user experience (Waytz et al., 2014). In other words, equipping virtual agents with natural language and oral communication methods that are similar to those of humans makes them more acceptable to users. This includes giving virtual agents personality traits such as emotions, preferences, and specific thinking patterns (Afzal et al., 2019; Bawack et al., 2021; Zhou et al., 2019). As described by Cohen et al. (2004), it is natural for us to infer the personality of voices during human-to-human conversations, and we do so even if what we hear is pre-recorded and played through a virtual agent. Thus, in terms of psychology, even if users know that a virtual agent consists of a series of algorithms, they still respond positively to humanlike traits such as emotions, way of talking, or preferred vocabulary (Short, 2017). The development of personality thus influences the building of trust, which is a crucial factor influencing the usage of smart services (Garcia & Lopez, 2018). Compared to technologies such as blockchain, the diffusion of smart services based on semantic technology proceeds relatively slowly (Sun & Li, 2021; Viriyasitavat et al., 2022). Personalizing services to build trust could be an important route towards a better adoption rate (Bedué & Fritzsche, 2021). Thus, a major challenge in VA development is understanding how the personality of virtual agents influences willingness to use VAs. At present, VA-related research has mostly focused on technology (e.g., enhanced speech recognition abilities), security (e.g., increased privacy protection and personal information security), and user behavior (e.g., factors that encourage VA usage). Little research has been conducted on how VA personality influences user trust, perceived risk, and willingness to use. With personal VAs as the context, the authors therefore examined the influence of VA personality (i.e., humanlike traits and behavior traits) on perceived risk, perceived enjoyment, trust, attitude towards use, and continued usage intention.

RELATED WORKS

VA interactions are achieved via a voice-activated user interface. Users need only give simple voice commands, from which VAs extract key information via semantic analysis, carry out the task, and then report the results. VAs are currently widely applied in banking, tourism, and e-commerce. Research involving VAs has so far focused on the following three aspects.

Technology

These studies have focused on strengthening VA capabilities (e.g., understanding human speech, producing voices similar to that of a real person, and adding emotional elements to the voices). For instance, Sermet and Demir (2021) introduced an open-source semantic web framework, Instant Expert, to help non-technical domain experts to build VAs for any web platform regardless of the underlying domain and technology. To improve semantic parsing, which is an important natural language processing (NLP) problem for VAs, Rongali et al. (2022) presented an automated methodology that utilizes modest amounts of unannotated data, which are typically easy to obtain. Kawase et al. (2020) employed a genetic algorithm to generate context-dependent parameters to enhance the accuracy of speech recognition. Other studies on enhancing VA capabilities include those by Park et al. (2018) and Sun (2020).

Security

This research has focused on detecting potential privacy and security issues and proposing solutions to enhance user trust. VAs have better “hearing” than human ears, and AI systems can detect many sounds that the human ear cannot. Yan et al. (2019) discovered that hackers may use ultrasonic sound waves that humans cannot perceive to issue malicious commands to VAs (e.g., Alexa and Microsoft Cortana) and steal personal information. Thapliyal et al. (2018) employed the Internet of Things and Amazon Echo to develop a VA system suitable for home use; this system has been verified as secure and expandable.

User Behavior

These studies have applied theoretical models from praxeology, psychology, and management to investigate the factors that influence willingness to use VAs (De Barcelos Silva et al., 2020). These studies focus on identifying important factors that influence why humans use VAs. For instance, McLean and Osei-Frimpong (2019) applied uses-and-gratification theory to find that personal and symbolic benefits are crucial factors underlying VA use, whereas the impact of hedonic benefits is small. Aside from personal benefits, the studies conducted by Nguyen et al. (2019) and Yang and Lee (2019) indicated that trust, perceived risk, perceived enjoyment, mobile self-efficacy, and perceived usefulness exert significant influence on willingness to use VAs.

In conclusion, existing studies have focused on in-depth investigations regarding aspects of technology, security, and user behavior. In terms of user behavior, the emphasis has been laid on factors influencing willingness to use VAs; little research has been conducted on human-computer interactions and VA personality. Nevertheless, an increasing number of studies have pointed out that personality will be a crucial direction for smart service development. VAs differ from chatbots, which simulate conversations using text messages. VAs interface through oral conversations, which are loaded with intuitive expressions and information. Importantly, voices contain a greater abundance of social cues than text, including personality, identity (e.g., gender and age), and emotions (e.g., amicability, happiness, and anger). On the whole, voice-based conversations offer a significant new direction for human-machine interactions.

RESEARCH MODEL AND HYPOTHESES

This study focused on the influence of VA personality on perceived risk, perceived enjoyment, trust, attitude to use, and continued usage intention, in which humanlike and behavior traits are considered the two crucial characteristics of VA personality. Figure 1 displays our research model, and Table 1 presents the operational definitions of our research constructs.

Figure 1. Conceptual model and hypotheses

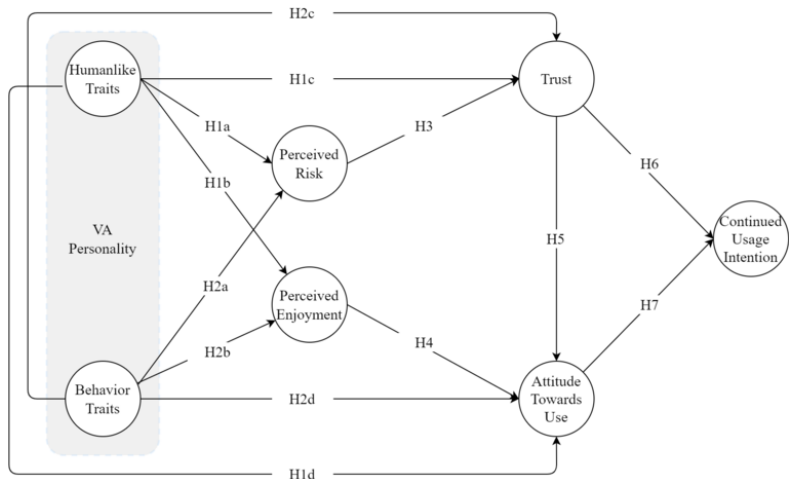


Table 1. Operational definitions

Constructs	Operational definitions	Items	References
Humanlike traits (HT)	Degree to which a VA seems like a real person during its interactions with users based on the characteristics of its voice (e.g., pitch, softness, wording, and tone).	5	Lewis and Hardzinski (2015)
Behavioral traits (BT)	Degree to which a VA is perceived to be unique (e.g., practical, informed, up-to-date, well-mannered, or helpful) during its interactions with users based on its responses.	5	Garcia and Lopez (2018)
Perceived risk (PR)	Uncertainty and potential adverse consequences that users perceive and the losses they subjectively expect when using a VA, including financial, performance-related, physical, mental, social, and time risks.	4	Xu and Gupta (2009)
Perceived enjoyment (PE)	Degree of enjoyment, positive emotions, and contentment that users experience during their interactions with a VA.	3	Lee et al. (2005)
Trust (TR)	Positive expectations that users have regarding the words, deeds, and promises of a VA and their belief that the VA and its provider will offer services in the best interests of the users rather than utilize their data for other purposes.	5	Dwyer et al. (2007) Metzger (2004)
Attitude towards use (AT)	Appraisal and reactions that users have towards a VA based on their own knowledge, emotions, and user experience.	3	Xu et al. (2013)
Continued usage intention (CU)	Subjectively assessed probability that a user will take a certain action, reflecting an individual's willingness to continue using VA services.	3	Ahn et al. (2007)

Hypothesis Development

Anthropomorphism refers to the attribution of qualities that are unique to humans (e.g., emotions, language, and actions) to non-human entities (e.g., pets) (Epley et al., 2007). People often display affection towards anthropomorphized entities (e.g., regarding a pet as a child and thus dressing and pampering it), and through these interactions, meet their own needs (e.g., companionship or desire for control) (Eyssel & Reich, 2013). From a commercial perspective, anthropomorphizing a product

is a profitable strategy. VAs are an artificially intelligent service software whose voice outputs are humanized to reap the benefits of anthropomorphism. VA and human interactions feature many of the characteristics common in human-to-human interactions. Research has shown that when users perceive similarity between themselves and a non-human entity, they will rely more heavily on their knowledge of self to infer, understand, and make psychological projections (DiSalvo et al., 2002). Eyssel and Kuchenbrandt (2012) pointed out that as humans have a far better understanding of themselves than non-humans (e.g., voicebots), they perceive a higher level of risk associated with unfamiliar products (e.g., a product with a low degree of anthropomorphism). This psychological mechanism is a form of self-protection. In contrast, if a non-human entity is perceived as similar to humans, users will not be as wary of it and will be more willing to use it and share their private thoughts with it (Kiesler et al., 2008). Similarity with humans is not limited to appearance; any similarity in behavior or even social relationships can have a comparable effect. VAs imitate natural human pronunciation and tone, allowing them to “talk” with users. Talking is a crucial means of emotional exchange in human society, and only with enough communication can there be mutual understanding, trust, and enjoyment perceived during interactions. In conclusion, the authors formulated the following hypotheses:

H1a: The humanlike traits of VAs exert a negative impact on perceived risk.

H1b: The humanlike traits of VAs exert a positive impact on perceived enjoyment.

H1c: The humanlike traits of VAs exert a positive impact on trust.

H1d: The humanlike traits of VAs exert a positive impact on attitude towards use.

Aside from the degree of anthropomorphism, the behavioral traits that users perceive during their conversations with the VAs may also exert a crucial influence. Research has shown that behavioral traits such as hand gestures, voice tones, and language are important social cues that people use to make inferences on the gender or personality of others to generate specific psychological impressions (e.g., like or dislike) (Isbister & Nass, 2000; Lee et al., 2006). In this study, the authors employed the widely used Google Assistant and Apple’s Siri as experiment tools. According to a study that surveyed 3,661 individuals between the ages of 18 and 65 in six countries (i.e., Argentina, Brazil, Chile, Spain, Germany, and the United Kingdom), Google Assistant and Apple’s Siri possess seven significant behavioral traits: practical, informed, up-to-date, well-mannered, logical, helpful, and kind (Garcia & Lopez, 2018). These behavioral traits are recognized as positive in human society. Research has indicated that in most cases, positive behavioral traits can significantly reduce the risk perceived by consumers and increase their willingness to use new technology. In interpersonal interactions, continued positive interactions also build a certain degree of trust and enjoyment. The authors therefore infer that the behavioral traits that VAs display during their interactions with users, including response speed, friendliness of wording, and quality of provided information, influence perceived risk, perceived enjoyment, trust, and attitude towards use. For example, Google Assistant allows users to use their usual vocabulary to make requests by speaking. Compared with inputting commands with a keyboard, using voice to conduct dialogue is more intuitive and convenient, which can effectively reduce user anxiety (Moreno & Mayer, 2007; Qiu & Benbasat, 2009). Based on the above friendly interaction model, users can quickly understand the benefits that applications such as VAs can bring to them; this can strengthen users’ trust in the system as well as their enjoyment in using it (Cheng et al., 2022; Natarajan & Gombolay, 2020). The authors therefore formulated the following hypotheses:

H2a: The behavioral traits of VAs exert a negative impact on perceived risk.

H2b: The behavioral traits of VAs exert a positive impact on perceived enjoyment.

H2c: The behavioral traits of VAs exert a positive impact on trust.

H2d: The behavioral traits of VAs exert a positive impact on attitude towards use.

Perceived risk refers to the unpredictable results or potential adverse circumstances that consumers perceive when purchasing a product or service; it is a subjective expected loss. Perceived risk generally has two important aspects: decision uncertainty (e.g., means of assessing the quality of a used car) and consequences of wrong decisions (e.g., whether the car will cause a family member to be injured because it is poor quality) (Featherman & Pavlou, 2003; Mitchell, 1999). In management, marketing, and consumer behavior, perceived risk has been found to be negatively correlated with product acceptance and trust and may include financial, psychological, social, private, safety, functional, and performance-related aspects (Bansal & Voyer, 2000; Benamati et al., 2021; Featherman & Pavlou, 2003; Park et al., 2019). VAs are a new type of technology, and most people are unfamiliar with the computational mechanisms of VAs, particularly with regard to personal privacy. When users feel that using VAs may pose significant risks, their trust in VAs will decline. The authors therefore formulated our third hypothesis as follows:

H3: Perceived risk exerts a negative influence on trust in VAs.

Perceived enjoyment is the degree of enjoyment that users experience during interactions with products or services. Researchers have indicated that intrinsic motivation plays a crucial role in the use of new technology. When users experience greater enjoyment, they will generate a positive attitude (Rouibah et al., 2021). Kim et al. (2013) discovered a positive relationship between perceived enjoyment and the quality of information system design: when users perceive that using a system is more interesting, they will rate its quality higher. The authors therefore propose our fourth hypothesis as follows:

H4: Perceived enjoyment exerts a positive influence on attitude towards using VAs.

Trust is an abstract and complex concept. Trust has been defined as “confidence in or reliance on some quality or attribute of a person or thing, or the truth of a statement” (McAllister et al., 2006; Simpson & Weiner, 2021). Within the context of technology adoption, Gefen (2002) explained trust as one party’s confidence that another party’s actions will meet their expectations. Specifically, the user trusts that the technical product is predictable, capable of accomplishing specified tasks, and acts with goodwill (McKnight et al., 2009). VAs are computer software (also known as a virtual agent) that receives and interprets instructions from the user and then retrieves the corresponding data from the database and feeds the data back to the user. If the database is not updated, the user will get the same answer by asking the same question. Thus, the predictability of VAs is high. Current mainstream VAs are able to complete tasks such as software function setting (e.g., alarm clock), schedule arrangement, information query (e.g., exchange rate), navigation, and purchasing goods (e.g., buying household supplies). They thus meet the needs of users in many daily activities. In terms of goodwill, as described previously, VAs such as Google Assistant and Apple’s Siri present well-mannered, helpful, and friendly personalities. Therefore, current mainstream VAs meet the conditions for trust. The positive personality traits of virtual agents in particular have been found to have a profound impact on consumer trust (Garcia & Lopez, 2018). The friendly and intuitive design of VAs means that users can quickly understand the benefits that VAs can bring to them; this further increases user trust in VAs as well as their enjoyment in using them (Cheng et al., 2022; Natarajan & Gombolay, 2020). Previous studies on online shopping have pointed out that trust is the foundation of the relationship between buyers and sellers. Basically, purchase intention is proportional to customer trust (Qalati et al., 2021; Zhao et al., 2020). Thus, in implementing innovative applications, such as mobile banking or e-government, building a foundation of trust is crucial to ensuring usage intentions (Lonkani et al., 2020; Rouibah et al., 2022). In other words, when users’ trust in VAs increases, their attitude towards using VAs should be more positive, and it will be easier for them to continue using VAs. Based on this discourse, the authors propose the fifth and sixth hypotheses as follows:

H5: Trust exerts a positive influence on attitude towards using VAs.

H6: Trust exerts a positive influence on intention to continue using VAs.

Attitude towards use refers to an individual's subjectively assessed probability that a certain action will be taken, reflecting the individual's willingness to adopt a certain product or service. Generally, an individual that has greater trust in a product or service will have a more positive attitude towards use. The technology acceptance model, which has widely been verified as highly reliable and valid, indicates that when people take a more positive attitude towards new technology, it will lead to a stronger intention to use said technology (Marangunić & Granić, 2015). Furthermore, attitude towards use has been demonstrated to be a crucial predictor of continued usage intention (Marangunić & Granić, 2015). In other words, a more positive attitude towards VA use indicates a greater willingness to continue to use VAs. The authors therefore formulated the following hypothesis:

H7: Attitude towards use of VAs exerts a positive influence on continued usage intention.

RESEARCH METHODOLOGY

Research Design

This study adopted the stimulus-organism-response (SOR) model (Mehrabian & Russell, 1974) and the theory of reasoned action (TRA) (Fishbein, 1979) to investigate the influence of VA personality on consumer trust and attitude towards VA use. According to the SOR model and the TRA, consumer behavior is triggered by various stimuli that originate from internal psychological and physiological factors (e.g., attitude towards VA or perceived enjoyment), subjective norms (e.g., social pressure from others), and the external environment (e.g., VA service quality). These stimuli strengthen the purchase motivation of consumers. In online shopping, the content (e.g., information and advertising) provided by VAs and their method of expression (i.e., VA personality) can change the emotional and cognitive state of customers, which then influences their attitude or behavioral responses (Chan et al., 2017). According to the TRA, the attitude of an individual refers to their positive or negative feelings and assessments towards engaging in certain behaviors (Ajzen, 1985; Fishbein, 1979). The authors thus established the research model of this study as shown in Figure 1. The authors referred to relevant literature for question items to measure each construct and adjusted them to fit our context. The operational definitions and source references of each construct are displayed in Table 1. For VA personality (i.e., humanlike traits and behavior traits), The authors referred to the ten question items developed by Lewis and Hardzinski (2015) and Garcia and Lopez (2018). Next, based on the consumer behavior studies conducted by Xu and Gupta (2009) and Lee et al. (2005), The authors adjusted the contents of the questionnaires to fit the overall perceptions of service quality (i.e., perceived risk and perceived enjoyment) that consumers have during their interactions with VAs, deriving seven question items. For trust, which measures the degree of trust that users have in the information content conveyed by VAs, the authors referred to the five question items designed by Dwyer et al. (2007) and Metzger (2004). Finally, for attitude towards use and continued usage intention, the authors respectively used three question items from the work of Xu et al. (2013) and Ahn et al. (2007). The resulting questionnaire contained a total of 28 question items all measured on a five-point Likert scale (1=strong disagreement to 5= strong agreement). The authors invited five domain experts and 20 consumers with VA experiences to participate in a pretest. After they completed the questionnaire, the authors asked them to offer suggestions on the wording of the question items. Based on their suggestions, the authors revised any ambiguous wording and removed statements and question items that seemed too similar. The Appendix displays the revised questionnaire.

Data Collection

Conversational service models with VAs as the medium are increasingly popular. They are gradually changing how companies interact with customers, and have been applied to a variety of scenarios (e.g., shopping, ticketing, and financial management). The authors therefore adopted convenience sampling (Evans & Mathur, 2005) to administer our survey to consumers with experience in using VA services. The questionnaire was administered electronically via SurveyCake (www.surveycake.com) for six weeks from 28 June 2022 to 9 August 2022. A total of 378 questionnaires were completed. After removing invalid questionnaires (e.g., respondents who never or rarely used VAs), the authors obtained 305 valid questionnaires for analysis. This sample size is in line with the recommendations of Kock and Hadaya (2018) for partial least square structural equation modeling; that is “sample size should be greater than ten times the maximum number of inner or outer model links pointing at any latent variable in the model” (Kock & Hadaya, 2018). Our study has a maximum of five inner links (within the “trust” construct) and five outer links (between the constructs of “trust” and “attitude”). Table 2 presents the demographic data of the respondents. Male respondents accounted for 48.2% of the valid samples. The largest age group contained respondents between the ages of 21 and 25 (17.7%) and the second-largest groups were between the ages of 51 to 55, and 56 to 60 (both 14.4%). About half the respondents (49.2%) had master’s degrees or higher and the next largest group (46.2%) had bachelor’s degrees. With regard to VA usage experience, 80.7% of respondents had used Apple’s Siri, 56.7% had used Google Assistant, and 22.3% had used MI AI; the use of other VAs was less than 10%. Apple’s Siri was most frequently used (63%), followed by Google Assistant (29.5%), and only 7.5% frequently used other VAs. Because the survey was conducted in Taiwan, the distribution of VA use may reflect the first-mover advantage, localization, and the easy availability of Apple and Google in Taiwan. Regarding the frequency of VA use, 38% of people reported using a VA every day and 37.7% reported using a VA several times a week.

DATA ANALYSIS AND RESULTS

The authors adopted the partial least squares (PLS) method and conducted data analysis using SmartPLS 3.3.3. PLS is a form of structural equation modeling (SEM) for statistical analysis that is well-suited to the size of the sample used in this study (i.e., 305 samples). To achieve stability in the various construct estimates, the authors employed bootstrap resampling with a sample size of 5,000 for parameter estimation and inference.

Instrument Validation

First, the authors examined the stability and consistency of model reliability, adopting Cronbach’s α to evaluate the reliability of the constructs (see Appendix). The Cronbach’s α coefficients ranged from 0.777 to 0.881, and were all greater than the standard of 0.7 (Bagozzi & Yi, 1988). Composite reliability (CR) ranged from 0.845 to 0.926, indicating good internal consistency among the constructs. The authors then examined the convergent validity and discriminant validity of the model. The former indicates a high degree of correlation among the observed variables in a single construct, which can then be used to measure the target construct (Churchill Jr., 1979). The factor loadings of some of the question items (HLT1, HLT3, and T3) did not reach the threshold of 0.7 suggested by Hair et al. (2010). However, some researchers have recommended keeping factor loadings greater than 0.6 or between 0.4 and 0.7 for PLS (Chin, 1998; Hulland, 1999; Moores & Chang, 2006). The average variance extracted (AVE) of the constructs all reached the suggested 0.5 (Fornell & Larcker, 1981), thereby indicating good convergent validity. With regard to discriminant validity, the authors followed the suggestions of Fornell and Larcker (1981). That is, the authors checked whether the square root of the AVE of each construct was greater than the coefficients of correlation between constructs and whether all the ratios in the matrix of heterotrait-monotrait ratio (HTMT) were less than 0.85 (Henseler

Table 2. Demographic profile

Variable	Characteristics	Sample	Percentage
Gender	Male	147	48.2%
	Female	158	51.8%
Age	under 20	15	4.9%
	21–25	54	17.7%
	26–30	26	8.5%
	31–35	13	4.3%
	36–40	28	9.2%
	41–45	32	10.5%
	46–50	30	9.8%
	51–55	44	14.4%
	56–60	44	14.4%
	above 61	19	6.2%
Level of study	high school	14	4.6%
	undergraduate	141	46.2%
	postgraduate	150	49.2%
VA brand used (can choose more than one VA)	Apple's Siri	243	80.7%
	Google Assistant	173	56.7%
	MI AI	68	22.3%
	Baidu DuerOS	22	7.2%
	Samsung Bixby	20	6.6%
	Microsoft Cortana	11	3.6%
	Amazon Alexa	3	1%
VA used most frequently	Apple's Siri	192	63%
	Google Assistant	90	29.5%
	others	23	7.5%
VA usage frequency	more than 10 times per day	18	5.9%
	1 to 10 times per day	116	38.0%
	several times a week	115	37.7%
	several times a month	56	18.4%

et al., 2015; Kline, 2015). The results in Tables 3 and 4 confirm discriminant validity among the constructs. Therefore, the reliability and validity of the model in this study were confirmed.

Hypothesis Testing

The authors adopted PLS-SEM to verify the path framework proposed in this study and used the coefficient of determination (R^2) to determine the explanatory power of the research model (Pavlou & Fygenson, 2006). In addition, the authors adopted the standardized root mean square residual (SRMR) as the index of overall model fitness. The details are as follows. Table 5 and Figure 2 present the hypothesis testing results of this study. The authors discuss them in three points. First,

Table 3. Correlations among latent constructs and Fornell-Larcker criterion

Construct	HLT	BT	PR	PE	TR	AT	CU
HLT	<i>0.735</i>	-	-	-	-	-	-
BT	0.593***	<i>0.768</i>	-	-	-	-	-
PR	0.022	0.065	<i>0.878</i>	-	-	-	-
PE	0.556***	0.601***	-0.025	<i>0.847</i>	-	-	-
TR	0.643***	0.601***	-0.110	0.625***	<i>0.723</i>	-	-
AT	0.424***	0.537***	-0.061	0.577***	0.518***	<i>0.847</i>	-
CU	0.420***	0.430***	-0.117	0.533***	0.477***	0.761***	<i>0.898</i>

Note: The diagonal matrix value (in italics and bold) presents the square root of AVE; *** p<0.001.

Table 4. HTMT test results

	HLT	BT	PR	PE	TR	AT	CU
HLT	-	-	-	-	-	-	-
BT	0.721	-	-	-	-	-	-
PR	0.058	0.077	-	-	-	-	-
PE	0.670	0.721	0.048	-	-	-	-
TR	0.799	0.711	0.244	0.764	-	-	-
AT	0.602	0.645	0.073	0.690	0.602	-	-
CU	0.493	0.503	0.134	0.623	0.552	0.827	-

the results for H1 indicate that the humanlike traits of VAs have a significant and positive effect on perceived enjoyment ($\beta=0.308$, $p<0.001$) and Trust ($\beta=0.439$, $p<0.001$), but have no significant effect on perceived risk ($\beta=-0.026$, $p=0.753$) and attitude to use ($\beta=-0.017$, $p=0.816$) (H1a, H1d). Thus, on the whole, H1 is only partially supported (i.e., H1b and H1c are supported while H1a and H1d are not supported). Second, the results for H2 indicate that the behavioral traits of VAs have a positive influence on perceived enjoyment ($\beta=0.418$, $p<0.001$), trust ($\beta=0.350$, $p<0.001$), and attitude towards use ($\beta=0.242$, $p<0.01$), but have no significant effect on perceived risk ($\beta=0.080$, $p=0.298$). On the whole, H2 is only partially supported (i.e., H2b~H2d are supported while H2a is not supported, $\beta=0.080$, $p=0.298$). Based on these results, we can infer that if VAs have features and behaviors that are more similar to those of humans, users will perceive more enjoyment, feel more trust in the software, and display more positive attitudes towards VAs. Thus, the analysis results showed that the perceived risk and perceived enjoyment of users have a significant influence on their trust in VAs ($\beta=-0.142$, $p<0.01$) and their attitude towards using VAs ($\beta=0.330$, $p<0.001$), respectively. Furthermore, greater trust in VAs prompted users to have a more positive attitude towards using VAs ($\beta=0.177$, $p<0.05$) and significantly increased their continued usage intention ($\beta=0.113$, $p<0.05$). Finally, when users have a more positive attitude towards using VAs, their continued usage intention also increases significantly ($\beta=0.703$, $p<0.001$). In other words, H3~H7 are all statistically supported.

In addition to these direct effects, the authors also tested whether the humanlike traits and behavioral traits of VAs influence continued usage intention via perceived enjoyment, trust, and attitude towards use. The authors employed SmartPLS 3.3.3 to test for these indirect effects. The results showed that the total indirect effect of humanlike traits on continued usage intention was

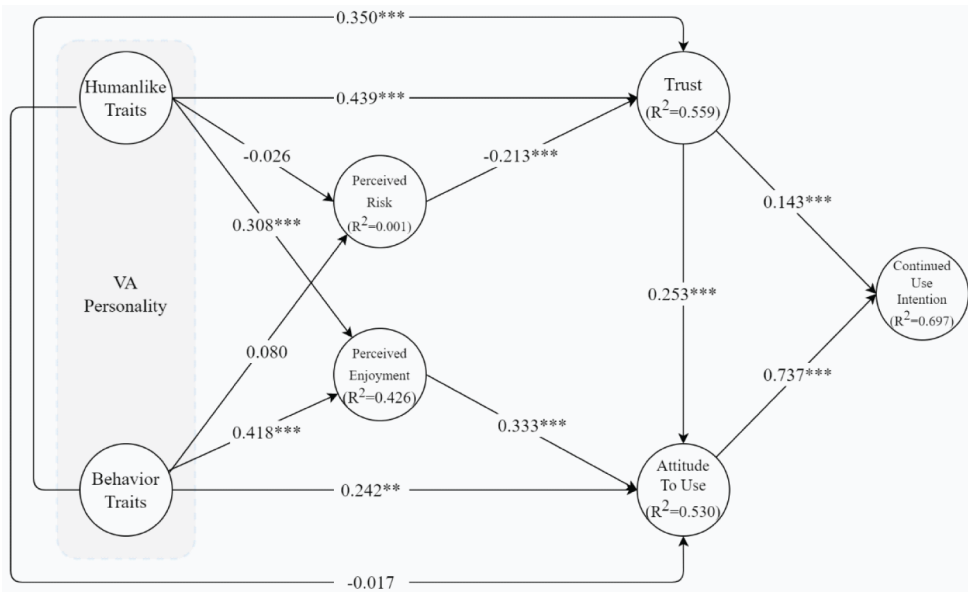
Table 5. Results of hypothesis-testing

Hypothesis		Path Coefficient	P Value	Findings
H1a	The humanlike traits of VAs exert a negative impact on perceived risk.	-0.026	0.753	Not Supported
H1b	The humanlike traits of VAs exert a positive impact on perceived enjoyment.	0.308	0.000	Supported
H1c	The humanlike traits of VAs exert a positive impact on trust.	0.439	0.000	Supported
H1d	The humanlike traits of VAs exert a positive impact on attitude towards use.	-0.017	0.816	Not Supported
H2a	The behavioral traits of VAs exert a negative impact on perceived risk.	0.080	0.298	Not Supported
H2b	The behavioral traits of VAs exert a positive impact on perceived enjoyment.	0.418	0.000	Supported
H2c	The behavioral traits of VAs exert a positive impact on trust.	0.350	0.000	Supported
H2d	The behavioral traits of VAs exert a positive impact on attitude towards use.	0.242	0.001	Supported
H3	Perceived risk exerts a negative influence on trust in VAs.	-0.142	0.002	Supported
H4	Perceived enjoyment exerts a positive influence on attitude towards using VAs.	0.330	0.000	Supported
H5	Trust exerts a positive influence on attitude towards using VAs.	0.177	0.014	Supported
H6	Trust exerts a positive influence on intention to continue to use VAs.	0.113	0.014	Supported
H7	Attitude towards use of VAs exerts a positive influence on continued usage intention.	0.703	0.000	Supported

0.165 ($p < 0.01$), whereas the total indirect effect of behavioral traits on continued usage intention was 0.348 ($p < 0.001$). In other words, the humanlike traits and behavioral traits of VAs indeed have a significant and positive impact on continued usage intention.

Regarding the overall explanatory power of the research model, Figure 2 presents the coefficient of determination (R^2) of the endogenous variables in the model. Except for the R^2 of perceived risk ($\beta = 0.005$, and neither HLT nor BT had any significant effect on it), the R^2 values of the constructs were all greater than 0.33. According to Chin (1998), $R^2 > 0.19$ indicates that the explanatory power of the model with regard to the endogenous variables is weak, whereas R^2 between 0.33 and 0.67 indicates moderate explanatory power. These standards indicate that the humanlike traits and behavioral traits of VAs have moderate explanatory power with regard to perceived enjoyment ($R^2 = 0.423$). Furthermore, the explanatory power of the model with regard to trust, attitude towards use, and continued usage intention is also moderate (R^2 ranges from 0.405 to 0.589). In conclusion, our research model offers a moderate degree of explanatory power regarding the various endogenous variables. Finally, in terms of model fit, the SRMR of the research model was 0.088. According to Garson (2016), an SRMR of less than 0.1 indicates a moderate degree of model fit. Thus, the research model in this study exhibits a moderate degree of model fit.

Figure 2. Structural model for main effects



DISCUSSION AND IMPLICATIONS

In this study, the authors developed an empirical approach based on the SOR model (Mehrabian and Russell, 1974) and the theory of reasoned action (Fishbein, 1979) to investigate how the personalities displayed by VAs stimulate consumers and generate varying degrees of changes in perceived risk, perceived enjoyment, attitude towards use, and trust, which in turn influence continued usage intention. In this section the authors discuss the empirical results, explain their contributions to theory and practice, and present the limitations of this study as well as directions for future research.

Discussion of Findings

First, the results of this study indicate that users derive more enjoyment from using VAs with more humanlike traits (e.g., appropriate tone and phrasing) and more positive behavioral traits (e.g., politeness and helpfulness). These factors also increase positive attitudes towards VA use. These results correspond with those of past studies (De Barcelos Silva et al., 2020). Perceived enjoyment is a crucial factor that determines whether users will ultimately continue using a product (Xu & Gupta, 2009). In consumer decisions, perceived enjoyment is a propelling force. When other conditions remain unchanged and the propelling force intensifies, the user will naturally be more willing to use a product or service. When individuals perceive enjoyment, their curiosity is aroused so that they continue interactions in the hope of deriving more satisfaction and pleasure. Therefore, individuals will be more willing to continue using VAs that satisfy their needs in enjoyable ways. In other words, VA personality reduces psychological impeding forces and strengthens propelling forces.

Second, regarding trust in VAs, the study results indicate that users have more trust in VAs with more humanlike and positive traits. This is an important finding for companies that are currently incorporating artificially intelligent voice-based services. At present, the reasoning mechanisms of a number of existing smart services are not disclosed, and they are often criticized as black-box operations, making users lose trust in smart services (e.g., concerns regarding data breaches and violations of personal rights); this is even more common among those who did not experience using

smart services in their childhoods (Shin, 2021; Weitz et al., 2019). Research has shown that to enhance user trust in smart services, users must be able to understand, explain, and authenticate the underlying voice-processing technologies and decision frameworks of the services (Pedreschi et al., 2019). For this reason, a number of researchers have attempted to enhance the interpretability of smart services from the perspective of technology and design specifications (Kawase et al., 2020; Sun, 2020). This study provides decision-makers with another perspective: using more humanlike designs and giving VAs personalities (e.g., adding emotions, preferences, and certain ways of thinking) can also build user trust. If the VA does not behave like a human, users will find it difficult to predict the VA's behavior based on their own experience. In contrast, VA anthropomorphism significantly reduces the psychological wariness of users. Importantly, this influence is intangible and does not have to be specifically explained to users. Successful anthropomorphism can significantly enhance user trust; this result should serve as good inspiration for corporate marketing and management decisions.

Third, the results of this study show that VA personality (i.e., humanlike traits and positive behavioral traits) has no significant effect on perceived risk. The authors believe that this may be because VA development is still in a stage of gradual growth, and some capabilities are immature. For example, although Apple's Siri and Google Assistant can perform many daily tasks, they currently have limited ability to carry out in-depth conversations with users. Thus, there is less effective communication between users and VAs, which means interaction with them is unlikely to significantly reduce perceived risk as suggested by some studies (Kiesler et al., 2008). As VAs develop, and interactions become more humanlike, there is a possibility that the cognitive risk perceived by users may decrease. Furthermore, this study mainly measures the perceived risk associated with the use of personal information by VAs. At present, the protection of personal data is highly valued around the world, especially in developed countries. This may also explain the insignificant relationship between VA personality and perceived risk.

Finally, based on the results of this study, the authors suggest that companies already providing or preparing to develop VA services consider strengthening the following VA traits. First, behavior traits (e.g., friendliness, politeness, and enthusiasm) and trending functions (e.g., interesting methods of feedback or the addition of popular terms) can be designed for VAs based on the characteristics and preferences of the target users. Next, the degree of VA anthropomorphism (e.g., introducing humanlike traits such as pitch, softness of the voice, wording, and tone) can be increased. These characteristics will make users put more trust in VA services and have more positive attitudes towards using VAs, which will then influence their continued usage intention. From a practical perspective, the global market of VAs is rapidly increasing, due to the ever-growing use of smartphones, intelligent home appliances, customer service automation systems, and in-car entertainment systems. In fact, this industry is expected to grow from \$13 billion in 2021 to \$245 billion in 2030. At present, North America is the region with the largest VA market, while Asia is the region with the fastest growth in VA services (Straits, 2022). Our empirical results in this study have notable implications for the promotion of VAs in various markets, particularly in service industries that depend on interactions with a large number of customers. In these situations, the personification of VAs could help to gain customer trust and convey a sense of warmth, thereby increasing customer willingness to continue using the service.

Limitations and Future Research

Despite our efforts to maintain high levels of rigor and consistency in the research design, data collection, analysis, and result interpretation, this study remains subject to the following limitations associated with time and resource factors: (1) The thinking patterns and psychological mechanisms that a personality entails are complex. This study merely performed analyses based on the anthropomorphic traits and behavior traits of VAs, which may not be comprehensive. Directions for future research include personality traits such as gender, emotions, and sense of security. (2) With regard to perceived risk, this study only measured the risk that individuals perceive regarding the VAs themselves and

did not include the information security awareness of the individuals, which may have caused some bias. Future studies could examine different risk factors, including psychological, social, and time risks. (3) This study only investigated two of the mainstream VAs (i.e., Google assistant and Apple's Siri) on the market. Therefore, our results may not be applicable to other VA services such as the voice services of self-driving cars. Future studies could pursue research models for other types of VA services and compare a range of services to further understand the influence of VA personalities. (4) Trust is bidirectional: the personification of VAs can enhance users' trust in them, and the higher the trust, the stronger their willingness to continue using VAs, while past studies have found that good user experiences enhance users' trust in innovative technologies (Cui et al., 2022; Dutton & Shepherd, 2006; Gefen, 2000). In other words, when users have a certain degree of trust in VA due to anthropomorphism, their willingness to use it will increase; if they continue to interact with the VA and gain positive experiences, this trust may be further strengthened, increasing their willingness to use it again, forming a positive cycle. This issue is worthy of further empirical research to better understand the relationship between trust and willingness to use.

CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

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REFERENCES

- Afzal, S., Dempsey, B., D'Helon, C., Mukhi, N., Pribic, M., Sickler, A., Strong, P., Vanchiswar, M., & Wilde, L. (2019). The personality of AI systems in education: Experiences with the Watson tutor, a one-on-one virtual tutoring system. *Childhood Education*, 95(1), 44–52. doi:10.1080/00094056.2019.1565809
- Ahn, T., Ryu, S., & Han, I. (2007). The impact of web quality and playfulness on user acceptance of online retailing. *Information & Management*, 44(3), 263–275. doi:10.1016/j.im.2006.12.008
- Ajzen, I. (1985). From intentions to actions: A theory of planned behavior. In J. Kuhl & J. Beckmann (Eds.), *Action control* (pp. 11–39). Springer. doi:10.1007/978-3-642-69746-3_2
- Al-Kaisi, A. N., Arkhangelskaya, A. L., & Rudenko-Morgun, O. I. (2021). The didactic potential of the voice assistant “Alice” for students of a foreign language at a university. *Education and Information Technologies*, 26(1), 715–732. doi:10.1007/s10639-020-10277-2
- Bagozzi, R. P., & Yi, Y. (1988). On the evaluation of structural equation models. *Journal of the Academy of Marketing Science*, 16(1), 74–94. doi:10.1007/BF02723327
- Bansal, H. S., & Voyer, P. A. (2000). Word-of-mouth processes within a services purchase decision context. *Journal of Service Research*, 3(2), 166–177. doi:10.1177/109467050032005
- Bawack, R. E., Wamba, S. F., & Carillo, K. D. A. (2021). Exploring the role of personality, trust, and privacy in customer experience performance during voice shopping: Evidence from SEM and fuzzy set qualitative comparative analysis. *International Journal of Information Management*, 58, 102309. doi:10.1016/j.ijinfomgt.2021.102309
- Bedué, P., & Fritzsche, A. (2021). Can we trust AI? An empirical investigation of trust requirements and guide to successful AI adoption. *Journal of Enterprise Information Management*, 35(2), 530–549. doi:10.1108/JEIM-06-2020-0233
- Benamati, J. H., Ozdemir, Z. D., & Smith, H. J. (2021). Information privacy, cultural values, and regulatory preferences. *Journal of Global Information Management*, 29(3), 131–164.
- Chan, T. K., Cheung, C. M., & Lee, Z. W. (2017). The state of online impulse-buying research: A literature analysis. *Information & Management*, 54(2), 204–217.
- Cheng, X., Zhang, X., Cohen, J., & Mou, J. (2022). Human vs. AI: Understanding the impact of anthropomorphism on consumer response to chatbots from the perspective of trust and relationship norms. *Information Processing & Management*, 59(3), 102940.
- Chin, W. W. (1998). Commentary: Issues and opinion on structural equation modeling. *Management Information Systems Quarterly*, 22(1), vii–xvi.
- Churchill, G. A. Jr. (1979). A paradigm for developing better measures of marketing constructs. *JMR, Journal of Marketing Research*, 16(1), 64–73.
- Cohen, M. H., Giangola, J. P., & Balogh, J. (2004). *Voice user interface design*. Addison-Wesley Professional.
- Cui, Y., Zhu, J., & Liu, Y. (2022). Exploring the social and systemic influencing factors of mobile short video applications on the consumer urge to buy impulsively. *Journal of Global Information Management*, 30(1), 1–23.
- De Barcelos Silva, A., Gomes, M. M., Da Costa, C. A., Da Rosa Righi, R., Barbosa, J. L. V., Pessin, G., De Doncker, G., & Federizzi, G. (2020). Intelligent personal assistants: A systematic literature review. *Expert Systems with Applications*, 147, 113193.
- DiSalvo, C. F., Gemperle, F., Forlizzi, J., & Kiesler, S. (2002, June 25). *All robots are not created equal: The design and perception of humanoid robot heads* [Conference paper]. Symposium on Designing Interactive Systems.
- Du, X., Zhao, X., Wu, C.-H., & Feng, K. (2021). Functionality, emotion, and acceptance of artificial intelligence virtual assistants: The moderating effect of social norms. *Journal of Global Information Management*, 30(7), 1–21.
- Dutton, W. H., & Shepherd, A. (2006). Trust in the Internet as an experience technology. *Information Communication and Society*, 9(4), 433–451.

- Dwyer, C., Hiltz, S., & Passerini, K. (2007, December). *Trust and privacy concern within social networking sites: A comparison of Facebook and MySpace* [Conference paper]. Americas Conference on Information Systems (AMCIS).
- Epley, N., Waytz, A., & Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4), 864–886. PMID:17907867
- Evans, J. R., & Mathur, A. (2005). The value of online surveys. *Internet Research*, 15(2), 195–219.
- Eyssel, F., & Kuchenbrandt, D. (2012). Social categorization of social robots: Anthropomorphism as a function of robot group membership. *British Journal of Social Psychology*, 51(4), 724–731. PMID:22103234
- Eyssel, F., & Reich, N. (2013, March 3–6). *Loneliness makes the heart grow fonder (of robots) – On the effects of loneliness on psychological anthropomorphism* [Conference paper]. 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI), Tokyo, Japan.
- Featherman, M. S., & Pavlou, P. A. (2003). Predicting e-services adoption: A perceived risk facets perspective. *International Journal of Human-Computer Studies*, 59(4), 451–474.
- Fernandes, T., & Oliveira, E. (2021). Understanding consumers' acceptance of automated technologies in service encounters: Drivers of digital voice assistants adoption. *Journal of Business Research*, 122, 180–191.
- Fishbein, M. (1979). A theory of reasoned action: Some applications and implications. *Nebraska Symposium on Motivation*, 27, 65–116. PMID:7242751
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *JMR, Journal of Marketing Research*, 18(1), 39–50.
- Funder, D. C. (2012). Accurate personality judgment. *Current Directions in Psychological Science*, 21(3), 177–182.
- Garcia, M. P., & Lopez, S. S. (2018, May 25–27). *Building trust between users and telecommunications data driven virtual assistants* [Conference paper]. 14th IFIP International Conference on Artificial Intelligence Applications and Innovations (AIAI), Rhodes, Greece.
- Garson, G. D. (2016). *Partial least squares regression and structural equation models* (2016 ed.). Statistical Publishing Associates.
- Gefen, D. (2000). E-commerce: The role of familiarity and trust. *Omega*, 28(6), 725–737.
- Gefen, D. (2002). Reflections on the dimensions of trust and trustworthiness among online consumers. *ACM SIGMIS Database: The DATABASE for Advances in Information Systems*, 33(3), 38–53.
- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. (2010). *Multivariate data analysis*. Pearson Education.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135.
- Holst, A. (2021). Smart home devices with voice assistants transaction value 2020–2025. *Statista*. <https://www.statista.com/statistics/1186644/smart-home-devices-voice-assistant-transactions/>
- Hulland, J. (1999). Use of partial least squares (PLS) in strategic management research: A review of four recent studies. *Strategic Management Journal*, 20(2), 195–204.
- Isbister, K., & Nass, C. (2000). Consistency of personality in interactive characters: Verbal cues, non-verbal cues, and user characteristics. *International Journal of Human-Computer Studies*, 53(2), 251–267.
- Kawase, T., Okamoto, M., Fukutomi, T., & Takahashi, Y. (2020). Speech enhancement parameter adjustment to maximize accuracy of automatic speech recognition. *IEEE Transactions on Consumer Electronics*, 66(2), 125–133.
- Ki, C. W. C., Cho, E., & Lee, J. E. (2020). Can an intelligent personal assistant (IPA) be your friend? Para-friendship development mechanism between IPAs and their users. *Computers in Human Behavior*, 111(106412).

- Kiesler, S., Powers, A., Fussell, S. R., & Torrey, C. (2008). Anthropomorphic interactions with a robot and robot-like agent. *Social Cognition*, 26(2), 169–181.
- Kim, J., Ahn, K., & Chung, N. (2013). Examining the factors affecting perceived enjoyment and usage intention of ubiquitous tour information services: A service quality perspective. *Asia Pacific Journal of Tourism Research*, 18(6), 598–617.
- Kline, R. B. (2015). Principles and practice of structural equation modeling. Academic Press.
- Kock, N., & Hadaya, P. (2018). Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods. *Information Systems Journal*, 28(1), 227–261.
- Laricchia, F. (2022). Number of digital voice assistants in use worldwide 2019–2024. *Statista*. <https://www.statista.com/statistics/973815/worldwide-digital-voice-assistant-in-use/>
- Lee, K. M., Peng, W., Jin, S.-A., & Yan, C. (2006). Can robots manifest personality?: An empirical test of personality recognition, social responses, and social presence in human–robot interaction. *Journal of Communication*, 56(4), 754–772.
- Lee, M. K., Cheung, C. M., & Chen, Z. (2005). Acceptance of internet-based learning medium: The role of extrinsic and intrinsic motivation. *Information & Management*, 42(8), 1095–1104.
- Lewis, J. R., & Hardzinski, M. L. (2015). Investigating the psychometric properties of the Speech User Interface Service Quality questionnaire. *International Journal of Speech Technology*, 18(3), 479–487.
- Lonkani, R., Changchit, C., Klaus, T., & Sampet, J. (2020). A comparative study of trust in mobile banking: An analysis of US and Thai customers. *Journal of Global Information Management*, 28(4), 95–119.
- Ma, Y., Drewes, H., & Butz, A. (2022, April 13–14). *How should voice assistants deal with users' emotions?* [Conference paper]. CHI'22 Workshop: The Future of Emotion in Human-Computer Interaction, New Orleans, LA.
- Marangunić, N., & Granić, A. (2015). Technology acceptance model: A literature review from 1986 to 2013. *Universal Access in the Information Society*, 14(1), 81–95.
- Martin, J., Mortimer, G., & Andrews, L. (2015). Re-examining online customer experience to include purchase frequency and perceived risk. *Journal of Retailing and Consumer Services*, 25, 81–95.
- McAllister, D. J., Lewicki, R. J., & Chaturvedi, S. (2006). Trust in developing relationships: From theory to measurement. *Academy of Management Proceedings*, 2006(1). 10.5465/ambpp.2006.22897235
- McCormick, R. (2017). Please don't make me talk to voice assistants anymore. *The Verge*. <https://www.theverge.com/2017/6/6/15744106/voice-assistants-siri-dont-make-me-talk>
- McKnight, H., Carter, M., & Clay, P. (2009). Trust in technology: Development of a set of constructs and measures. *DIGIT 2009 Proceedings*. <https://aisel.aisnet.org/cgi/viewcontent.cgi?article=1009&context=digit2009>
- McLean, G., & Osei-Frimpong, K. (2019). Hey Alexa... examine the variables influencing the use of artificial intelligent in-home voice assistants. *Computers in Human Behavior*, 99, 28–37.
- Mehrabian, A., & Russell, J. A. (1974). *An approach to environmental psychology*. MIT Press.
- Metzger, M. J. (2004). Privacy, trust, and disclosure: Exploring barriers to electronic commerce. *Journal of Computer-Mediated Communication*, 9(4).
- Mitchell, V. W. (1999). Consumer perceived risk: Conceptualisations and models. *European Journal of Marketing*, 33, 163–195.
- Moffat, D. (1997). Personality parameters and programs. In *Creating personalities for synthetic actors* (pp. 120–165). Springer.
- Moore, T. T., & Chang, J. C.-J. (2006). Ethical decision making in software piracy: Initial development and test of a four-component model. *Management Information Systems Quarterly*, 30(1), 167–180.
- Moreno, R., & Mayer, R. (2007). Interactive multimodal learning environments. *Educational Psychology Review*, 19(3), 309–326.

- Namahoot, C. S., Brückner, M., & Sivilai, S. (2021, September 25–28). *Smart, practical, and low-cost assistant system for hospital nutritionists in times of a pandemic* [Conference paper]. International Conference on Cooperative Design, Visualization and Engineering, Krakow, Poland.
- Natarajan, M., & Gombolay, M. (2020, March 23–26). *Effects of anthropomorphism and accountability on trust in human robot interaction* [Conference paper]. ACM/IEEE International Conference on Human-Robot Interaction, Cambridge, UK.
- Nguyen, Q. N., Ta, A., & Prybutok, V. (2019). An integrated model of voice-user interface continuance intention: The gender effect. *International Journal of Human-Computer Interaction*, 35(15), 1362–1377.
- Park, J., Amendah, E., Lee, Y., & Hyun, H. (2019). M-payment service: Interplay of perceived risk, benefit, and trust in service adoption. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 29(1), 31–43.
- Park, J., Son, H., Lee, J., & Choi, J. (2018). Driving assistant companion with voice interface using long short-term memory networks. *IEEE Transactions on Industrial Informatics*, 15(1), 582–590.
- Pavlou, P. A., & Fygenon, M. (2006). Understanding and predicting electronic commerce adoption: An extension of the theory of planned behavior. *Management Information Systems Quarterly*, 30(1), 115–143.
- Pedreschi, D., Giannotti, F., Guidotti, R., Monreale, A., Ruggieri, S., & Turini, F. (2019, January 27–February 1). *Meaningful explanations of Black Box AI decision systems* [Conference paper]. 33rd AAAI Conference on Artificial Intelligence, Honolulu, HI.
- Purington, A., Taft, J. G., Sannon, S., Bazarova, N. N., & Taylor, S. H. (2017, May 6–11). “*Alexa is my new BFF*”: Social roles, user satisfaction, and personification of the Amazon Echo [Conference paper]. CHI Conference Extended Abstracts on Human Factors in Computing Systems, Denver, CO.
- Qalati, S. A., Vela, E. G., Li, W., Dakhan, S. A., Hong Thuy, T. T., & Merani, S. H. (2021). Effects of perceived service quality, website quality, and reputation on purchase intention: The mediating and moderating roles of trust and perceived risk in online shopping. *Cogent Business & Management*, 8(1), 1869363.
- Qiu, L., & Benbasat, I. (2009). Evaluating anthropomorphic product recommendation agents: A social relationship perspective to designing information systems. *Journal of Management Information Systems*, 25(4), 145–182.
- Rongali, S., Arkoudas, K., Rubino, M., & Hamza, W. (2022, July 23–29). *Training naturalized semantic parsers with very little data* [Conference paper]. Thirty-First International Joint Conference on Artificial Intelligence, Vienna, Austria.
- Rouibah, K., Al-Qirim, N., Hwang, Y., & Pouri, S. G. (2021). The determinants of eWoM in social commerce: The role of perceived value, perceived enjoyment, trust, risks, and satisfaction. *Journal of Global Information Management*, 29(3), 75–102.
- Rouibah, K., Qurban, H., & Al-Qirim, N. (2022). Impact of risk perceptions and user trust on intention to re-use e-government: A mixed method research. *Journal of Global Information Management*, 30(1), 1–29.
- Rzepka, C. (2019, August). *Examining the use of voice assistants: A value-focused thinking approach* [Conference paper]. Twenty-fifth Americas Conference on Information Systems, Cancun, Mexico.
- Schwartz, E. H. (2019). 6 reasons consumers don’t own smart speakers, and which ones matter. *Voicebot.ai*. <https://voicebot.ai/2019/06/26/6-reasons-consumers-dont-own-smart-speakers-and-which-ones-matter/>
- Sciuto, A., Saini, A., Forlizzi, J., & Hong, J. I. (2018, June 9–13). “*Hey Alexa, what’s up?*”: A mixed-methods studies of in-home conversational agent usage [Conference paper]. Designing Interactive Systems Conference, New York, NY.
- Sermet, Y., & Demir, I. (2021). A semantic web framework for automated smart assistants: A case study for public health. *Big Data and Cognitive Computing*, 5(4), 57.
- Shin, D. (2021). The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI. *International Journal of Human-Computer Studies*, 146, 102551.
- Short, T. (2017, October 5–9). *Designing stronger AI personalities* [Conference paper]. AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment, Snowbird, Little Cottonwood Canyon, UT.

- Simpson, J., & Weiner, E. (2021). *Oxford English dictionary*. Clarendon Press.
- Straits. (2022). *Intelligent virtual assistant market*. <https://straitsresearch.com/report/intelligent-virtual-assistant-market>
- Sun, T. W. (2020). End-to-end speech emotion recognition with gender information. *IEEE Access: Practical Innovations, Open Solutions*, 8, 152423–152438.
- Sun, Y., & Li, Y. (2021). The impact of risk-aware consumer trust on CB e-commerce platforms and purchase intention. *Journal of Global Information Management*, 30(3), 1–13.
- Thapliyal, H., Ratajczak, N., Wendroth, O., & Labrado, C. (2018, December 17–19). *Amazon echo enabled IoT home security system for smart home environment* [Conference paper]. IEEE International Symposium on Smart Electronic Systems (iSES) (Formerly iNiS), Hyderabad, India.
- Viriyasitavat, W., Xu, L. D., Sapsomboon, A., Dhiman, G., & Hoonsopon, D. (2022). Building trust of blockchain-based Internet-of-Thing services using public key infrastructure. *Enterprise Information Systems*, 16(12), 1736–1759.
- Waytz, A., Heafner, J., & Epley, N. (2014). The mind in the machine: Anthropomorphism increases trust in an autonomous vehicle. *Journal of Experimental Social Psychology*, 52, 113–117.
- Weitz, K., Schiller, D., Schlagowski, R., Huber, T., & André, E. (2019, July 2–5). “Do you trust me?” *Increasing user-trust by integrating virtual agents in explainable AI interaction design* [Conference paper]. 19th ACM International Conference on Intelligent Virtual Agents, Paris, France.
- Xu, H., & Gupta, S. (2009). The effects of privacy concerns and personal innovativeness on potential and experienced customers’ adoption of location-based services. *Electronic Markets*, 19(2–3), 137–149.
- Xu, J., Benbasat, I., & Cenfetelli, R. T. (2013). Integrating service quality with system and information quality: An empirical test in the e-service context. *Management Information Systems Quarterly*, 37(3), 777–794.
- Yan, C., Zhang, G., Ji, X., Zhang, T., Zhang, T., & Xu, W. (2019). The feasibility of injecting inaudible voice commands to voice assistants. *IEEE Transactions on Dependable and Secure Computing*.
- Yang, H., & Lee, H. (2019). Understanding user behavior of virtual personal assistant devices. *Information Systems and e-Business Management*, 17(1), 65–87.
- Zhao, S., Fang, Y., Zhang, W., & Jiang, H. (2020). Trust, perceived benefit, and purchase intention in C2C e-commerce: An empirical examination in China. *Journal of Global Information Management*, 28(1), 121–141.
- Zhou, M. X., Mark, G., Li, J., & Yang, H. (2019). Trusting virtual agents: The effect of personality. *ACM Transactions on Interactive Intelligent Systems*, 9(2–3), 1–36.

APPENDIX

Table 6.

Construct	Item	Question	Factor loading	CR	AVE
Humanlike traits Cronbach's α : 0.786	HLT1	The VA uses terminology I am familiar with.	0.673	0.854	0.540
	HLT2	The VA sounds logical.	0.706		
	HLT3	The VA sounds like the voices I hear on the radio or on television.	0.675		
	HLT4	The VA sounds like a normal person.	0.788		
	HLT5	The VA's voice sounds natural.	0.820		
Behavior traits Cronbach's α : 0.826	BT1	The VA seems practical.	0.766	0.878	0.589
	BT2	The VA seems well-informed.	0.798		
	BT3	The VA seems up-to-date.	0.775		
	BT4	The VA seems well-mannered.	0.728		
	BT5	The VA seems kind.	0.770		
Perceived risk Cronbach's α : 0.853	PR 1	I feel that my personal information may be misused by the VA system.	0.871	0.910	0.771
	PR 2	When using a VA, I think that my personal information may be accessed by unauthorized personnel.	0.880		
	PR 3	I am worried that someone with ill intentions may eavesdrop on my private conversations with the VA.	0.883		
Perceived enjoyment Cronbach's α : 0.803	PE1	I think that using VA systems is interesting.	0.721	0.883	0.718
	PE2	The process of using a VA is enjoyable.	0.909		
	PE3	I have fun while using a VA.	0.899		
Trust Cronbach's α : 0.777	T1	I trust that when I am using a VA, my personal information will only be used by the VA vendor to complete the task I assigned.	0.756	0.845	0.522
	T2	I trust that when I am using a VA, my personal information is protected.	0.706		
	T3	When I am interacting with a VA, I can trust the information it provides to me.	0.675		
	T4	I think that the talking style displayed by the VA is trustworthy.	0.788		
	T5	I trust the VA because when I talk with it, it feels like I am talking with a normal person.	0.820		
Attitude towards use Cronbach's α : 0.845	A1	On the whole, I believe that VAs are worth using.	0.868	0.907	0.764
	A2	I think that using VAs is a good idea.	0.883		
	A3	On the whole, my feelings regarding the use of VAs are positive.	0.871		
Continued usage intention Cronbach's α : 0.881	CU1	I plan to continue using VAs.	0.908	0.926	0.807
	CU2	I plan to use VAs frequently in the future.	0.899		
	CU3	I will recommend VAs to other people.	0.888		

Note: CR: composite reliability, AVE: average variance extracted

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